Remote Sensing (SUR 4380)

Spring 2026 | Online Asynchronous | 3 Credits

Instructor Information

Instructor: Dr. Eben Broadbent

School of Forest, Fisheries, and Geomatics Sciences

Spatial Ecology & Conservation Lab – www.speclab.org

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Office Hours: Tuesdays 10:30 am – 12:30 pm (in-person or Zoom) and by appointment. Students may request alternate times if circumstances require.

Course Overview

Course Description

This course introduces scientific and engineering principles of remote sensing, including electromagnetic radiation, optical and thermal imaging, satellite and airborne platforms, LiDAR and radar systems, preprocessing, image classification, cloud-based analysis, artificial intelligence, and applied environmental and geospatial applications. The course emphasizes conceptual understanding, hands-on exploration of real datasets, troubleshooting software installations, and the development of technical competency across multiple remote-sensing platforms.

Prerequisites

Senior standing in the Geomatics major (UF catalog requirement). Other students may enroll with instructor's permission.

Course Purpose

The course combines foundational science with practical, experiential exercises. Students learn remote-sensing workflows, including data acquisition, preprocessing, classification, accuracy assessment, LiDAR interpretation, radar analysis, and Google Earth Engine (GEE) scripting.

Learning Objectives

Upon successful completion of this course, students will be able to:

1. Explain electromagnetic radiation principles as applied to remote sensing.
2. Describe modern passive and active sensors and their applications.
3. Pre- and post-process satellite, aerial, drone, and LiDAR data using ENVI and ArcGIS Pro and other software.
4. Interpret multispectral, hyperspectral, thermal, radar, and LiDAR datasets.
5. Conduct image classifications and evaluate accuracy.
6. Apply cloud-based analytical tools, including Google Earth Engine.
7. Communicate results through maps, figures, tables, and scientific writing.

Topics Covered

Flight Platforms: Space (satellites, space station), high-altitude (balloons), aircraft (fixed-wing, helicopters), drones (fixed-wing, VTOL, copter), and terrestrial systems.

Passive Remote Sensing

Electromagnetic Spectrum Fundamentals: Energy sources, radiation principles, atmospheric interactions, Earth surface interactions.

Visual Imaging: Photography elements, film types (B&W, color, color infrared), camera formats, digital imaging, storage and manipulation, applications in topographic/planimetric mapping and environmental monitoring.

Image Processing: Geometric correction, orthorectification, photogrammetry, digital elevation models (DEMs), digital surface models (DSMs).

Multi- and Hyperspectral Imaging: Platforms, preprocessing (radiometric correction, radiance to reflectance, atmospheric models), spatial and spectral enhancement, pan-sharpening.

Classification: Unsupervised (ISODATA), supervised (nearest neighbor), accuracy assessment (error matrices), applications in vegetation and mineralogy.

Thermal Infrared Imaging: Thermal radiation principles, radiant vs. kinetic temperature, applications in conservation biology, human presence monitoring, plant health, and fire dynamics.

Active Remote Sensing

Radar: Satellite and aircraft-borne systems, resolution (azimuth, range), terrestrial systems, ground-penetrating radar.

LiDAR: Applications in terrain analysis, pollution detection, moisture assessment, and target recognition.

Data Fusion: Active + passive integration, NEON Airborne Observation Platform (AOP), GatorEye Unmanned Flying Laboratory, segmentation (eCognition).

Cloud-Based Analysis: Google Earth Engine (GEE).

Artificial Intelligence: Machine learning and deep learning applications in remote sensing, including pattern and context extraction from imagery, automated species identification, and AI-assisted classification workflows.

Course Materials

Required Textbooks

Remote Sensing and Image Interpretation (Lillesand, Kiefer & Chipman, 7th ed.)

Remote Sensing of Vegetation (Jones & Vaughan, 1st ed.)

Optional Textbook

Remote Sensing: Principles, Interpretation, and Applications (Sabins & Ellis, 4th ed.)

Software

ENVI, ArcGIS Pro (via UFApps or student license), and additional software for drone and LiDAR workflows.

Software Access: The preferred method for accessing ArcGIS Pro and ENVI is through UF Apps (info.apps.ufl.edu), which ensures consistent access regardless of operating system. Students may install locally for faster performance. When using UF Apps, data cannot be

stored directly on your computer—follow provided instructions for managing files within the UF Apps storage system. Students who wish to install ENVI locally may request a student license. Detailed setup guidance will be provided in Module 1, Week 1 materials and lab recordings on Canvas.

Course Website

Canvas (all course materials, announcements, and submissions)

Technology Requirements

Hardware & Connectivity

- Computer capable of handling large geospatial datasets
- Broadband internet (wired connection required for quizzes and exams)
- Webcam and microphone for synchronous sessions and office hours
- Updated browser (Canvas supports only the two most recent versions)

Required Skills

Ability to manage files, install software, and use geospatial tools.

Proctoring & Synchronous Meetings

Honorlock proctoring software (with required browser extensions) is required for exams. Synchronous meetings, including office hours and Q&A sessions, will be conducted through Zoom via links provided in Canvas. Review Zoom's system requirements to ensure compatibility.

Library Access (Off-Campus)

You must have the UF VPN Client installed and be logged in before viewing UF Library eResources or course reserve materials from off-campus. See How to Install Cisco AnyConnect for UF VPN. For technical assistance, call the UF Computing Help Desk at 352-392-4357.

Digital Information Literacy

Students will evaluate data quality, interpret metadata, locate and assess scientific articles, and understand dataset provenance.

Academic Calendar & Schedule

UF Spring 2026 Key Dates

Classes Begin: January 12, 2026

Drop/Add Ends: January 16, 2026

MLK Day (No Classes): January 19, 2026

Spring Break: March 14–21, 2026

Classes End: May 1, 2026

Final Exam Window: May 2–7, 2026

Weekly Course Schedule

Subject to modifications—see the most updated schedule in the Pages section of Canvas.

Week 1: Introduction to course, EMR basics

Week 2: Platforms (satellite, aircraft, UAV)

Week 3: EMR interactions, atmospheric effects

Week 4: Ortho-rectification, photogrammetry

Week 5: Multispectral/hyperspectral imaging

Week 6: Hyperspectral interpretation

Week 7: Unsupervised classification (ISODATA)

Week 8: Supervised classification + accuracy assessment

Week 9: Thermal infrared imaging and artificial intelligence

Spring Break: No work

Week 10: Radar imaging, Sentinel-1, UAVSAR

Week 11: LiDAR, GEDI, NEON AOP, GatorEye

Week 12: Segmentation (eCognition, ENVI, SegmentAnything)

Week 13: Cloud computing with Google Earth Engine (GEE) and Earth AI

Week 14: Independent project development

Week 15: Final project submission and presentations

Final Exam: May 2–7 (or TBD)

Assignments & Grading

Grade Components

Lecture Questions & Activities – 20%

Includes Zoom polls during lectures (true/false, multiple choice, short response) and other graded forms to assess understanding and participation.

Lecture Intensive Projects – 30%

Start during lecture period and may extend beyond. Work on data acquisition, theory, analysis, and applications, with delivery via uploads and/or submitted presentations.

Independent Projects & Presentations – 20%

Intensive longer duration projects which are developed and presented. These are remote sensing-based studies resulting in a 5-minute presentation and 5-pg paper, using a standard format with sections: title, introduction, question/hypothesis, methods, results, discussion, conclusions. Must include at least one map, one table, one additional figure, and three peer-reviewed citations. Literature reviews are not acceptable, rather students must engage in addressing a research question using real remote sensing data and techniques learned in this class or elsewhere.

Exams (Midterm & Final) – 30%

30–60 questions (true/false, multiple choice, short response), completed in under 60 minutes using Honorlock. Closed-book: no internet, notes, books, or aids—memory only.

Grading Scale

A = 93%+ | A− = 90–92.99% | B+ = 87–89.99% | B = 83–86.99% | B− = 80–82.99%

C+ = 77–79.99% | C = 73–76.99% | C− = 70–72.99%

D+ = 67–69.99% | D = 63–66.99% | D− = 60–62.99% | F < 60%

Course Policies

Late Submissions & Make-Up Work

Students are responsible for accessing online lectures, readings, quizzes, and exams on time. Late tutorial worksheets incur a 3% penalty per day (up to 42% maximum) and will not be accepted more than two weeks after the due date. Lab reports and projects also cannot be submitted after two weeks. Quizzes cannot be submitted late; Canvas close dates are firm.

Acceptable reasons for make-up work follow UF attendance policy. Students must contact the instructor within 24 hours of any issue. Technical problems (other than UF e-Learning outages) do not excuse late work; requests for consideration must include a Help Desk ticket documenting the time and nature of the problem.

Academic Integrity & Plagiarism

You will use various resources to support your work. Failure to attribute credit to sources violates UF's Honor Code and carries serious consequences. You must properly cite: direct quotations, paraphrased or summarized ideas, and facts that are not common knowledge. When in doubt, cite it! This preserves academic integrity, shows your research extent, and allows readers to pursue topics further.

Follow instructor guidelines for citation formatting. For more information, visit the UF Library Guide: Attribution & Plagiarism.

Professional Conduct

Submissions displaying unprofessional or unconstructive behavior may receive an immediate zero. Unprofessional behavior during lectures or group activities will result in warnings and may result in zeros for associated activities. Repeated violations may result in removal from the course.

Netiquette (Online Communication)

General Guidelines:

- Treat everyone with respect in all communications
- Use clear, concise language with correct spelling and grammar
- Avoid slang and texting abbreviations
- Use standard fonts (e.g., Times New Roman, 12–14 pt)
- Avoid all caps (interpreted as yelling)
- Be cautious with humor/sarcasm—tone can be lost in text
- Protect personal information (yours and others')

Discussion Board Guidelines:

- Stay on-topic and within course scope
- Review and edit posts before sending
- Be thorough but concise; give proper credit to sources
- Read all messages in a thread before replying
- Add value—don't just repeat others' posts or say "I agree"
- Express disagreement respectfully; avoid personal remarks

Class Recordings

Students may record video or audio of class lectures for personal educational purposes, submitting a complaint, or use as evidence in legal proceedings. Recordings may not be published or shared without written instructor permission. "Class lecture" refers to instructor-delivered presentations (including instructor-led discussions) but excludes lab sessions, student presentations, clinical activities, assessments, field trips, or private conversations. Unauthorized publication may result in civil action and/or disciplinary measures under UF Regulation 4.040.

University Policies

Academic policies for this course are consistent with university policies. See https://syllabus.ufl.edu/syllabus-policy/uf-syllabus-policy-links/

Campus Health and Wellness Resources

Visit https://one.uf.edu/whole-gator/topics for resources that are designed to help you thrive physically, mentally, and emotionally at UF.
Please contact UMatterWeCare for additional and immediate support.

Software Use

All faculty, staff and students of the university are required and expected to obey the laws and legal agreements governing software use. Failure to do so can lead to monetary damages and/or criminal penalties for the individual violator. Because such violations are also against university policies and rules, disciplinary action will be taken as appropriate.

Privacy and Accessibility Policies

[required for online courses, list all technology used]

- Instructure (Canvas)
  - Instructure Privacy Policy
  - Instructure Accessibility
- Zoom
  - Zoom Privacy Policy
  - Zoom Accessibility

Learning Strategies

- Save readings and handouts to your own computer—courses close about a week after semester end and you cannot be re-added to retrieve materials.
- Use the UF Writing Studio for writing support (online appointments available).
- Explore LinkedIn Learning (log in via the eLearning page) for free tutorials on software and skills.
- Refrain from using the Canvas mobile app for submitting assignments or exams.